SAR Lab 7: Xinmo Landslide Analysis

Event: Xinmo landslide, Sichuan Province, China. Event date: 2017-06-24. AOI: 103.62 to 103.68 E, 32.04 to 32.09 N. LiCSAR frame: 062D_05831_131313.

1. Introduction and Study Objective

This report recreates the SAR Lab 7 Xinmo landslide workflow using public COMET-LiCSAR interferograms and MintPy SBAS processing. It then compares LiCSAR coherence, Sentinel-1 GRD VV change, and Sentinel-2 optical changes for event mapping. The objective is to evaluate what can be recovered from public pre-processed interferograms and open Sentinel data, not to exactly reproduce the supplied PS-based study.

2. Data and Study Area

The study area is centered near 103.6506 E, 32.0661 N in Sichuan Province, China. The working AOI is 103.62, 32.04, 103.68, 32.09. The SAR time-series analysis uses LiCSAR frame 062D_05831_131313. Event mapping uses an event-spanning LiCSAR coherence pair, Sentinel-1 GRD VV images, and Sentinel-2 optical imagery.

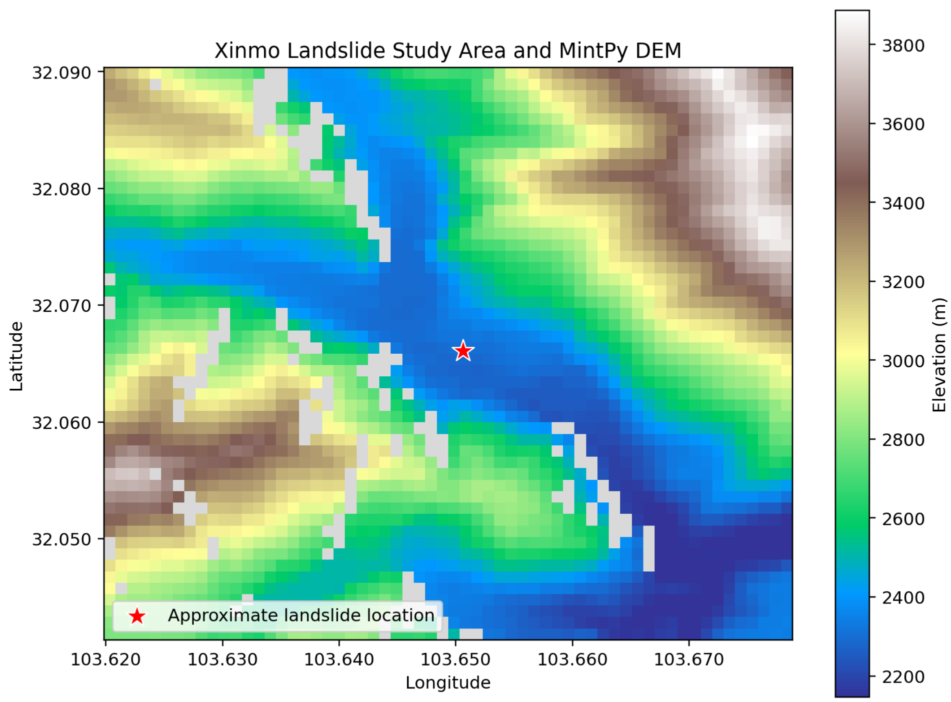

Figure 1. Study area and topographic context for the Xinmo landslide AOI. Source: lab-generated DEM/context figure; coordinates are longitude and latitude.

3. LiCSAR and MintPy SBAS Methodology

Pre-event LiCSAR unwrapped phase and coherence products were downloaded for interferograms before 2017-06-24. A 72-day temporal-baseline selection produced a disconnected network, so the maximum temporal baseline was expanded to 108 days. The final MintPy input stack contains 42 acquisitions and 127 interferograms from 2014-10-09 to 2017-06-07. The perpendicular baseline range is approximately -178 m to +189 m. The LiCSAR products were converted to MintPy HDF5 inputs, subset to the AOI, and processed with coherence-weighted SBAS inversion.

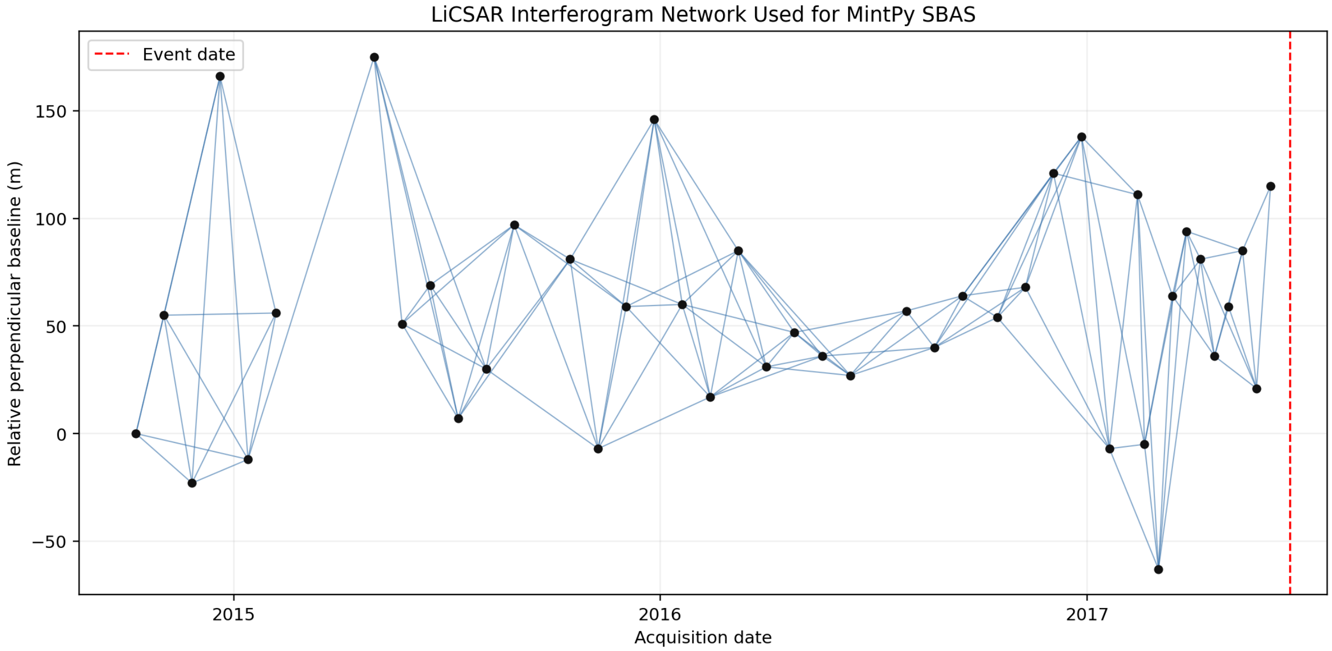

Figure 2. MintPy interferogram network for the pre-event LiCSAR stack. The network is connected and spans 2014-10-09 to 2017-06-07. Source: LiCSAR interferograms processed with MintPy.

4. Reference-Point Selection

The final run used a manual local reference point at Y/X = 28,31, lat/lon = 32.06233, 103.65083. This point is close to the landslide but outside the interpreted affected area and has high temporal coherence in the MintPy result. The manual reference reduces sensitivity to long-wavelength atmospheric, seasonal, or topographic differences. The tradeoff is that all displacement is relative to this selected pixel.

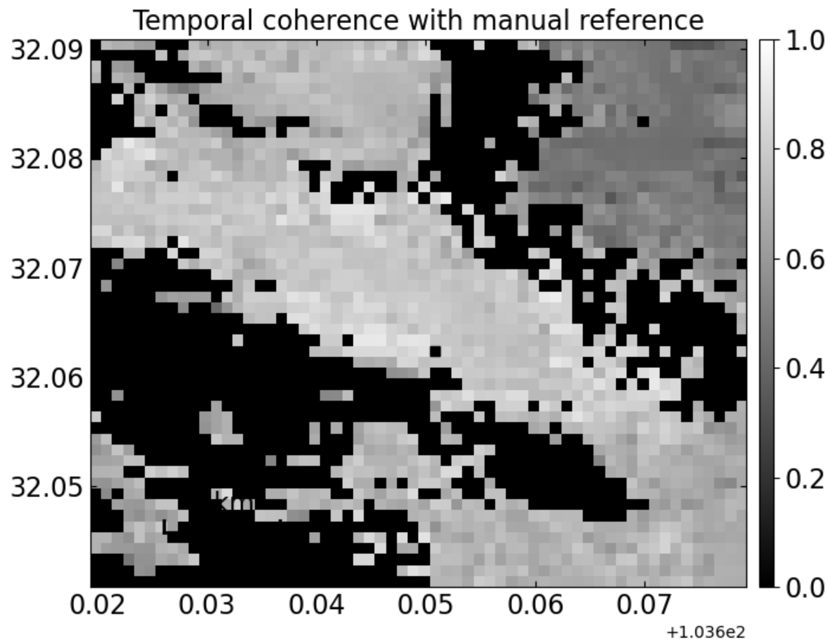

Figure 3. Manual reference context and temporal coherence. The selected reference point is local to the landslide and coherent in the MintPy result. Source: LiCSAR/MintPy temporal coherence.

5. Pre-Failure Time-Series Results

The representative landslide pixel at Y/X = 29,29, lat/lon = 32.06133, 103.64883 shows a broad negative LOS trend of about -1.07 +/- 0.08 cm/year. This supports pre-event motion at the landslide, but the available stack does not show a clear late-stage acceleration. The key limitation is acquisition timing: the stack ends on 2017-06-07, which is 17 days before the 2017-06-24 failure.

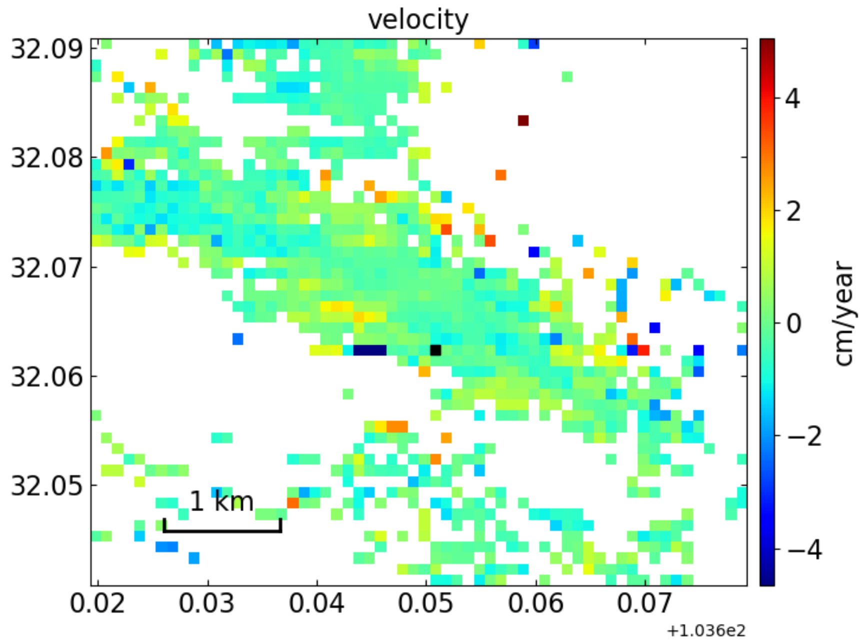

Figure 4. MintPy LOS velocity after manual rereferencing. Units are cm/year. Source: LiCSAR/MintPy SBAS velocity product.

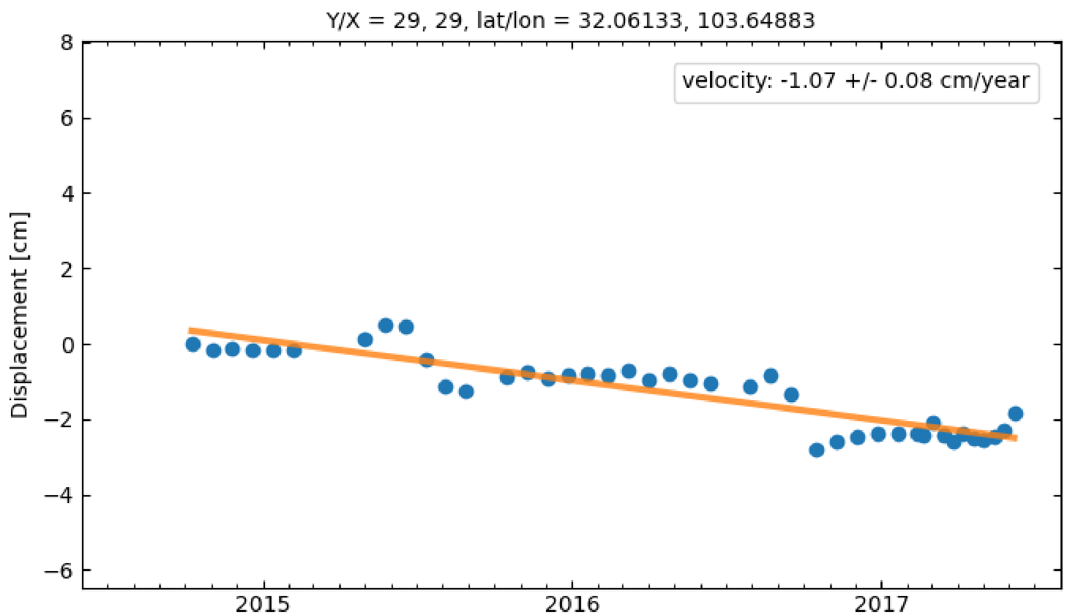

Figure 5. Representative landslide-pixel displacement time series after manual rereferencing. Units are cm of relative LOS displacement. Source: LiCSAR/MintPy time-series product.

6. Coherence Analysis

The selected event-spanning pair is 20170607_20170725 with a 48-day temporal baseline. The mean coherence in the landslide-centered sample is 0.046. The mean coherence at the manual-reference sample is 0.053. Both local samples are very low coherence, so this pair does not cleanly separate the landslide from nearby low-coherence terrain.

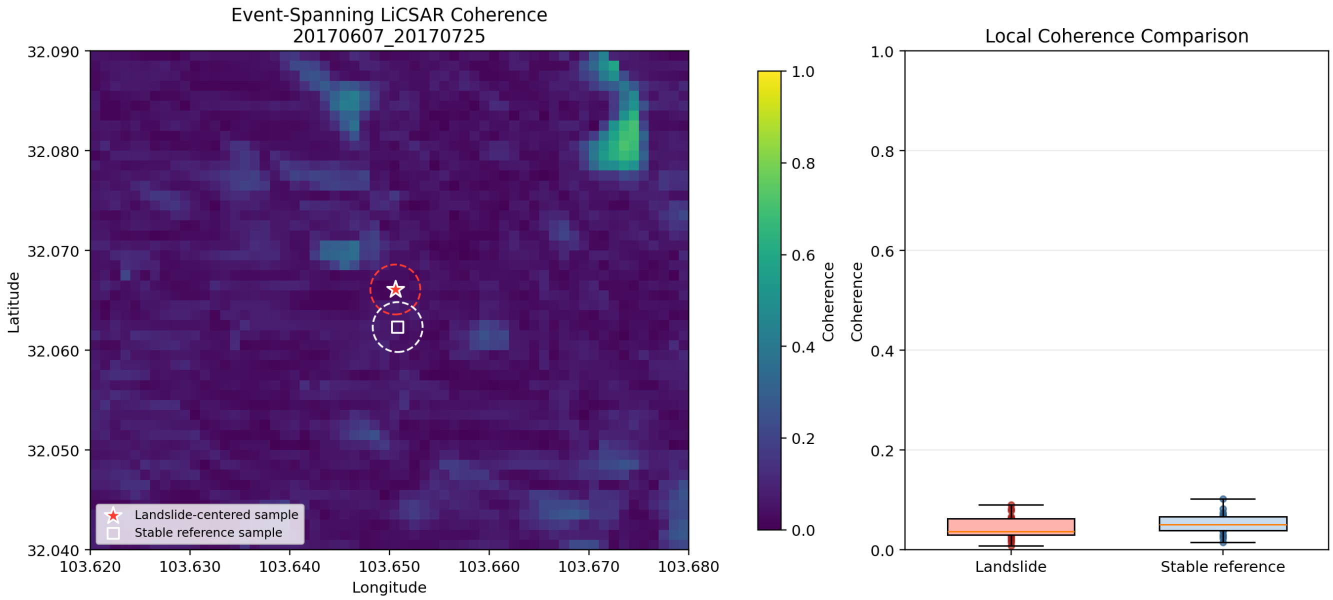

Figure 6. Event-spanning LiCSAR coherence for pair 20170607_20170725. Color scale is coherence from 0 to 1. Source: LiCSAR .geo.cc.tif product.

7. Sentinel-1 VV Analysis

The Sentinel-1 GRD comparison uses descending relative orbit 62. The pre-event acquisition is 2017-06-19 and the post-event acquisition is 2017-07-13. VV change is post-event VV minus pre-event VV in dB. The landslide-centered sample has mean VV change of -0.49 dB, while the stable comparison sample has mean VV change of -0.89 dB. The local contrast is small, so VV change is weak for mapping the landslide in this case.

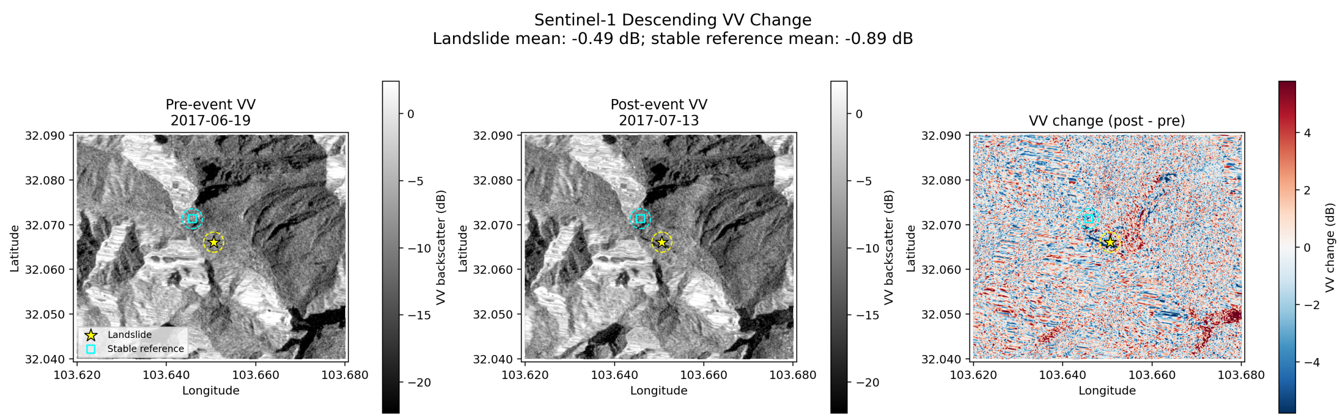

Figure 7. Sentinel-1 descending VV backscatter before and after the event, plus post-minus-pre VV change in dB. Source: Copernicus Sentinel-1 GRD via Google Earth Engine.

8. Sentinel-2 and Spectral-Index Analysis

Sentinel-2 surface-reflectance imagery did not provide a usable 2017 pre/post pair for this AOI, so the workflow used COPERNICUS/S2_HARMONIZED Level-1C top-of-atmosphere imagery. The pre-event date is 2017-02-19; the post-event date is 2017-08-13. The final affected-area mask uses post brightness >= 0.115, post NDVI <= 0.0, and NDVI change <= -0.10. The largest connected component inside a 1.8 km search radius is retained. The resulting affected area is 80.04 ha, or 0.8004 km2.

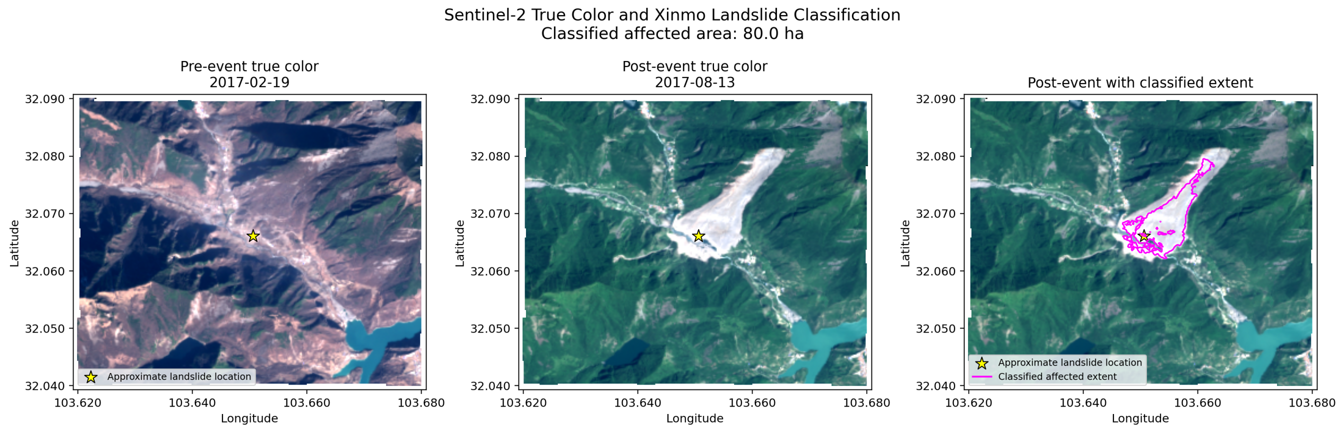

Figure 8. Sentinel-2 pre-event true color, post-event true color, and refined affected-area extent. Source: Copernicus Sentinel-2 Level-1C via Google Earth Engine.

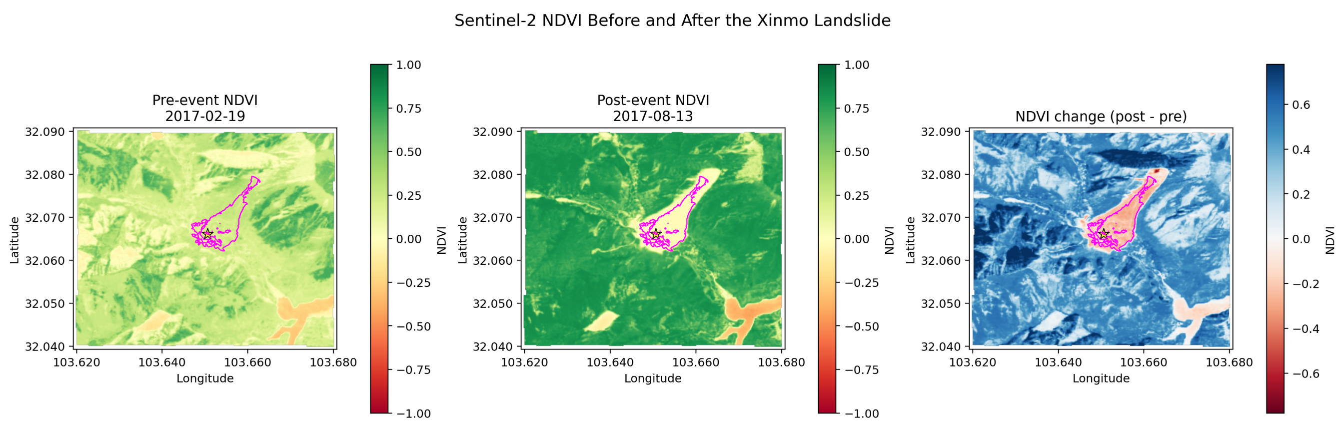

Figure 9. Sentinel-2 NDVI before, after, and post-minus-pre change. NDVI is unitless. Source: Copernicus Sentinel-2 via Google Earth Engine.

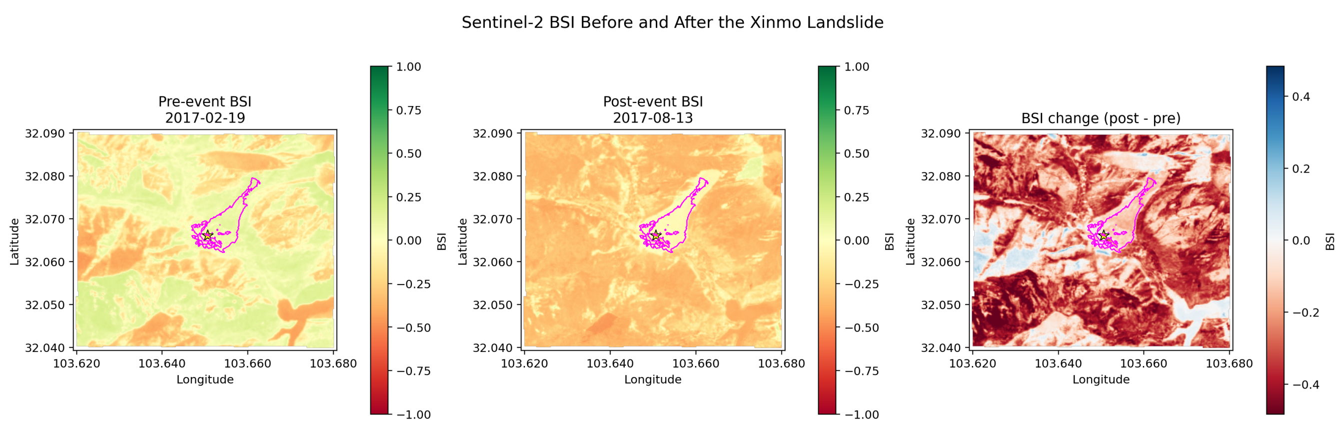

Figure 10. Sentinel-2 BSI before, after, and post-minus-pre change. BSI is unitless. Source: Copernicus Sentinel-2 via Google Earth Engine.

9. Comparison and Discussion

Method	Main signal	Result	Interpretation
MintPy SBAS	Pre-failure LOS motion	-1.07 +/- 0.08 cm/year	Useful for motion, not extent
LiCSAR coherence	Event disturbance	0.046 vs 0.053	Weak separator
Sentinel-1 VV	Backscatter change	-0.49 dB vs -0.89 dB	Weak contrast
Sentinel-2 true color	Optical scar	80.04 ha	Clearest extent
NDVI change	Vegetation loss	-0.229 vs 0.333	Strong support
BSI change	Bare soil/debris	-0.108 vs -0.221	Not reliable alone

Sentinel-2 true color plus NDVI decrease gives the clearest landslide extent. SBAS is useful for pre-failure deformation but cannot map the event extent. Coherence and Sentinel-1 VV are weaker in this workflow because their local contrasts are small or spatially ambiguous.

10. Conclusions

- MintPy SBAS supports pre-event motion at the landslide, with an inspected trend of about -1.07 +/- 0.08 cm/year.
- The manual reference point gives a local relative basis for displacement interpretation.
- The available LiCSAR stack ends on 2017-06-07, so late acceleration before the 2017-06-24 failure cannot be reproduced.
- Coherence loss is weak for this pair because both local samples have very low coherence.
- Sentinel-1 VV change is also weak because local backscatter contrast is small.
- Sentinel-2 true color plus NDVI decrease provides the clearest spatial map, with affected area 80.04 ha.
- NDVI change supports vegetation loss or burial; BSI change is not a reliable positive-threshold discriminator for this pair.

11. References

- SAR Lab 7 assignment handout, SAR_Lab_7.pdf.
- Supplied Xinmo landslide PS-based study: DOI 10.1007/s10346-017-0915-7.
- COMET-LiCSAR public interferogram products, frame 062D_05831_131313.
- Yunjun, Z., Fattahi, H., and Amelung, F. MintPy: a Python package for InSAR time series analysis.
- ESA Copernicus Sentinel-1 GRD imagery.
- ESA Copernicus Sentinel-2 imagery.
- Google Earth Engine cloud-processing platform.